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PARAMETER EFFICIENT AUDIO CAPTIONING WITH FAITHFUL GUIDANCE USING AUDIO-TEXT SHARED LATENT REPRESENTATION

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ABSTRACT

There has been significant research on developing pretrained transformer architectures for multimodal-to-text generation tasks. Albeit performance improvements, such models frequently suffer from hallucination and large memory footprint making them challenging to deploy on edge devices. In this paper, we address both these issues for the application of automated audio captioning. First, we propose a data augmentation technique for generating hallucinated audio captions and show that similarity based on an audio-text shared latent space is suitable for detecting hallucination. Then, we propose a parameter efficient inference time faithful decoding algorithm that enables smaller audio captioning models with performance equivalent to larger models trained with more data. During the beam decoding step, the smaller model utilizes an audio-text shared latent representation to semantically align the generated text with corresponding input audio. Faithful guidance is introduced into the beam probability by incorporating the cosine similarity between latent representation projections of greedy rolled out intermediate beams and audio clip. We show the efficacy of our algorithm on benchmark datasets and evaluate the proposed scheme against baselines using conventional audio captioning and semantic similarity metrics while illustrating tradeoffs between performance and complexity.

***Index Terms*—** Audio captioning, Hallucination, CLAP

1. INTRODUCTION

In recent years, there has been extensive research on pushing the boundaries of multimodal-to-text generation tasks like image captioning [1], audio captioning ([2], [3]) etc. Although there has been significant research in improving model performance, two major bottlenecks, hallucination [4] and large memory footprint[5], remain that inhibit the wide scale adoption of such tasks on constrained computing devices. In [4], hallucination is defined as ”the generated content that is nonsensical or unfaithful to the provided source content”. Their survey documents research on hallucination in multimodal to text generation tasks such as abstractive summarization and vision-language generation. Second, improved performance of large pretrained transformer models on evaluation benchmarks comes at the cost of larger memory footprint[5]. These models are often over-parameterized with respect to the task they are solving, thus necessitating architectural innovations to tackle computational complexity during deployment.

In this paper, we address both of these bottlenecks by proposing automated audio captioning models retrofitted with a hallucination detection and mitigation at decoding stage. Audio captioning is the task of generating a descriptive caption given an audio clip. Audio tags are audio-event labels associated with a given audio clip. To the best of our knowledge, we are the first to make the following contributions in this domain. First, we propose a data augmentation technique to generate hallucinated audio captions using existing audio captioning datasets by leveraging large language models. Second, we provide an intuitive reasoning on why existing audio captioning metrics are not suitable for detecting hallucination. Instead we argue that acoustic similarity of captions needs to be taken into account and introduce a hallucination metric based on an audio-text shared latent representation. Third, we propose an inference time hallucination mitigation algorithm where similarity of the intermediate beams with respect to the input audio is used to faithfully guide the beams during beam decoding. We show that our retrofitting method enables smaller audio captioning models with performance equivalent to much larger models.

2. RELATED WORKS

2.1. Audio Captioning

Conventional audio captioning systems are based on encoder-decoder architectures where the encoder captures the temporal and acoustic information of the audio and the decoder generates the caption auto-regressively([3]). In addition to the encoded audio representations, keywords are extracted from the audio and provided to the decoder to achieve grounded guidance([2], [3]). The study on lexical diversity and similarity of captions[6] shows that different annotators interpret the same audio using different vocabulary. In this work, we analyze the properties of SOTA audio captioning evaluation metrics when used for semantic and acoustic similarity detection and find that similarity metrics based on audio-text shared latent representation are better suited for such tasks.

Original Caption	Injecting Audio Tags	Hallucinated Caption
A campfire in the night time with crickets and other bugs making noise in the background.	Bird, Speech, Outside, urban or manmade	A nighttime campfire with crickets and other bugs chirping in the background, accompanied by the sound of human speech.
A crowd of people and a child begin talking as cars beep in the background and then the crowd cheers.	Outside, urban or manmade, Singing, Insect	A child is playing outside in an urban area while singing and insects are heard in the background.

Table 1: Hallucinated audio captions generated using the proposed data augmentation technique.

Data\Metrics	BLEU 1	METEOR	ROUGE L	CIDER	SPICE	SentBert	CLAPScore$_{tt}$
Hallucinated	0.4338	0.223	0.3773	0.2077	0.1491	0.5414	0.3609
Non Hallucinated	0.5798	0.3408	0.5663	0.8031	0.217	0.8627	0.701

Table 2: Performance of evaluation metrics on audio captioning hallucination benchmark.

2.2. Hallucination in natural language generation and its detection metrics

Mitigating hallucination via training[7] and during inference time([8], [9]) is an ongoing focus of natural language research. FactEdit[7] performs rewriting of the generated summary to avoid hallucination while [8], [9] propose decoding techniques to reduce hallucination on the fly during decoding time. In [1], hallucination is investigated in image captioning and a CHAIR metric is proposed that computes the ratio of generated objects to objects found in image and ground-truth captions. Further, it is observed in [10] that models that perform well on standard captioning metrics like CIDER[3] still produce unfaithful texts. In [11], the problem of hallucination in video captioning is studied. To our knowledge, we are the first to study hallucination in the audio captioning domain.

3. METHODOLOGY

We divide our proposed methodology into three sections. First, we explain our novel data augmentation technique to generate hallucinated audio captions. Second, we investigate and propose a hallucination metric to measure hallucinations in audio captioning. Third, we explain in detail our proposed faithful decoding algorithm.

3.1. Generating hallucinated data

In order to investigate and study hallucination in audio captioning, we introduce a data augmentation technique that removes zero or more audio events from the original caption and gradually augments it with similar/dissimilar audio events. Table 1 illustrates generated hallucinated captions using this method. First, we randomly select 50 examples from the Clotho[12] dataset. We randomly select one of the five ground truth captions as original caption and paraphrase it using vicuna checkpoint[13] of LLaMA 2[14] to generate the non hallucinated data points. Second, we retrieve audio tags using Audio spectrogram transformer[15] for the corresponding audio clip. We assume the audio tags with low classification scores are acoustically dissimilar from the input audio. So, we randomly select three tags in the range of 30-40(dissimilar) from a list of descending order ranked audio tags based on classification score. We generate a modified audio caption using LLaMA 2[14] by providing the original audio caption and audio tags. In addition, we append a few examples to the prompt to enable in-context learning by which the LLM learns correlation between input and output.

3.2. CLAPScore as hallucination metric

For a metric to be considered suitable for hallucination detection, it should have two properties - 1) detect false positive audio events in the caption 2) account for acoustic similarity between audio events. Contrastive Language Audio Pretraining (CLAP) [16] uses an audio and text encoder to learn a shared embedding space using contrastive learning which shows SOTA results for downstream tasks like audio retrieval, audio captioning [3]. We introduce CLAPScore to compute the cosine similarity over audio-text and text-text pairs as shown in Equation 1.

$$\text{CLAPScore} = \frac{x_A \cdot x_B}{|x_A| \cdot |x_B|} \tag{1}$$

Where x_A and x_B represents CLAP projected representations of audio or text. Hereon, we denote the CLAPScore between audio and text as CLAPScore$_{at}$ while between two texts as CLAPScore$_{tt}$.

We adopt the audio captioning metrics used by [3] including CIDER, SPICE, BLEU, METEOR, ROUGE L scores and text semantic similarity based metric SentBert[17]. For the first condition, we show the performance of standard metrics on the generated hallucinated and non-hallucinated, as described in section 3.1, serving as benchmark. For the second condition, acoustic similarity detection needs to be an inherent property of the metric. Since it is difficult to categorize audio clips into acoustically similar pairs systematically, we perform a qualitative study as shown in Table 3.

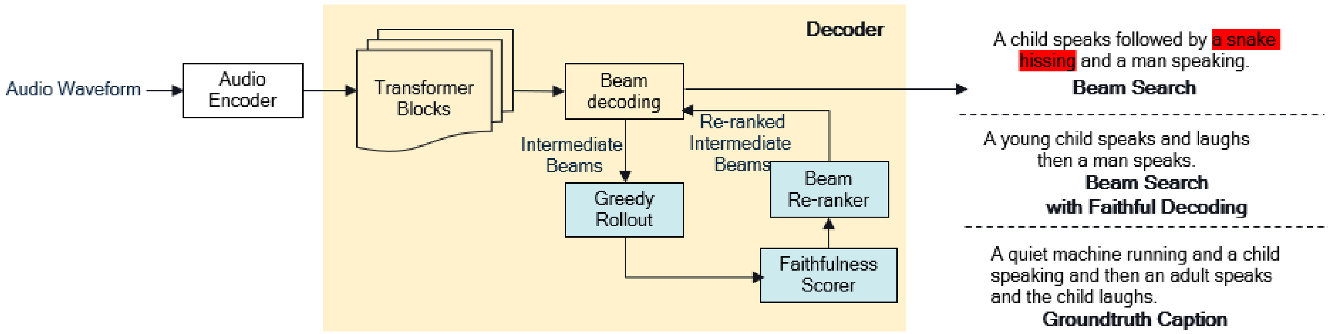

Fig. 1: Proposed faithful decoding for audio captioning. The proposed components are colored light sky blue.

Example pair\Metrics	BLEU-1	METEOR	ROUGE-L	CIDER	SPICE	SentBert	CLAP Score$_{tt}$
Horse is trotting., Someone is tapping on a surface.	0.3033	0.0702	0.193	0	0	-0.0572	0.6476
Horse is trotting., Someone is running on wood.	0.1947	0.0379	0.2179	0	0	0.2262	0.7223
Crowd is applauding the performer., Crows is silent after the performer's show.	0.439	0.2374	0.5908	0.0	0.4444	0.3596	0.3836

Table 3: Qualitative analysis of evaluation metrics on acoustic similar and dissimilar audio captions. The first two pairs are similar acoustic captions while the third pair is an example of dissimilar acoustic caption.

From Table 2 and 3, we can observe that none of the metrics except CLAPScore$_{tt}$ satisfy both these properties. Keeping 0.6 as threshold for discrimination, found by observation, CLAPScore$_{tt}$ classified hallucinated and non-hallucinated data with an accuracy of 92.3% and 82.14% respectively. Although the standard metrics perform well on detecting text hallucinations in Table 2, they are unable to distinguish between acoustic similar and dissimilar audio captions as shown in Table 3. This is due to them considering only text embedding space during computation. The audio captions - "Horse is trotting." and "Someone is walking on the wood." might seem very different in the text domain whereas in acoustic domain both sounds are acoustically similar to hear. Hence, such audio events should be less penalized by a hallucination metric compared to the acoustically contrasting audio caption pairs such as ("Crowd is applauding the performer.", "Crowd is silent after the performer's show"). Since CLAPScore$_{tt}$ satisfies both the properties, in this paper, we adopt CLAP as the hallucination metric.

3.3. Faithful decoding algorithm

In this section, we propose a faithful decoding algorithm that utilizes the ability of CLAP to detect hallucinations (shown in Section 3.2) during inference time.

3.3.1. Greedy rollout

Beam search performs a breadth first search at each decoding step with limited branches from Begin of sentence (BOS) to End of sentence (EOS) [18]. Each path from BOS to EOS are called hypothesis. During beam decoding process, only partial hypothesis or intermediate beams (paths that start at BOS and end before EOS) are available for re-ranking. To compare the intermediate beams against the input audio, we complete them using greedy search[18]. Greedy search samples the token with highest probability at every decoding step as shown in Equation 2. This serves us as a look ahead for how the beam would pan out in case it goes down that direction.

$$P_\Theta(y|x) = \prod_{t=1}^{|y|} P_\Theta(y_t|x, y_{<t}), \qquad (2)$$

where x is the input and y_t is the word generated at t^{th} decoding step.

3.3.2. Faithfulness scorer

Next, we compute the relevance of greedy rolled out beam with the input audio by taking the CLAP projections of beam text and audio. We normalize the projections and take cosine similarity to compute CLAPScore$_{at}$ as the distance between greedy rolled out beam and audio in the shared embedding space (Equation 1).

3.3.3. Beam re-ranker

To incorporate the CLAP$_{score}$ into beam decoding we weight it over the model probability P$_i$ to compute P$_{weighted}$ (Equation 3). The modified probability P$_{weighted}$ ensures to guide the beam to explore regions faithful to input audio thereby reducing hallucination.

$$\mathrm{P}_{weighted} = (1-\alpha)\mathrm{P}_i + \alpha \mathrm{CLAPScore}_{at} \qquad (3)$$

where P_i denotes the model probability for i^{ith} token.

Dataset+Sampling	BLEU 1	METEOR	ROUGE L	CIDER	SPICE	SentBert	CLAPScore$_{tt}$
AC+Beam	0.6515	0.2174	**0.4597**	0.6101	0.1597	0.7035	0.8208
AC+CLAP Beam	0.6323	0.2267	0.4455	0.6249	0.1595	0.7298	0.8242
AC+HTSAT-BERT Beam	**0.6605**	**0.2330**	0.4584	**0.6593**	**0.1677**	**0.7554**	**0.8618**
C+Beam	0.5496	0.1706	**0.3738**	0.3745	0.1141	0.6098	0.7404
C+CLAP Beam	0.4931	0.1711	0.3412	0.3184	0.1185	0.6405	0.7814
C+HTSAT-BERT Beam	**0.5570**	**0.1793**	0.3695	**0.4187**	**0.1283**	**0.6641**	**0.7995**

Table 4: Main Result: Performance of baseline with proposed faithful decoding on AudioCaps(AC) and Clotho(C) datasets. The bold numbers signify best performance on corresponding dataset.

Dataset+Sampling	CIDER	SPICE	SentBert	CLAP Score$_{tt}$
AC+Beam	**0.7904**	0.1821	**0.7926**	0.8849
AC+CLAP Beam	0.7252	0.1806	0.7847	0.8727
AC+HTSAT-BERT Beam	0.7232	**0.1828**	0.7915	**0.8859**
C+Beam	**0.4794**	0.1323	**0.6766**	**0.8175**
C+CLAP Beam	0.4394	0.1317	0.6735	0.8079
C+HTSAT-BERT Beam	0.4462	**0.1324**	0.6744	0.8154

Table 5: Performance of HTSAT-BART[3] with proposed faithful decoding on AudioCaps(AC) and Clotho(C) datasets. The bold numbers signify best performance on corresponding dataset.

4. EXPERIMENTS

4.1. Datasets

We trained the models on Clotho[12] and AudioCaps[19] audio captioning datasets. Clotho audio captioning dataset comprises of 4981 audio samples with each sample accompanied by 5 human written captions. The duration of the audio clips are in the range of 15 to 30 sec. Audiocaps[19] consist of 46k human written captions obtained via crowdsourcing with 10 sec duration for each audio clip. For both the datasets, we use the test set for our evaluation. For fixed length transformer encoders like HTSAT-BERT[3], we truncate the audio sample to 10 sec and resample at the rate of 32000.

4.2. Experiment Setup

We demonstrate the performance of our model against [20] chosen especially due to its small size. The model is trained on Clotho and AudioCaps datasets from scratch and evaluated correspondingly. It consists of CNN10 PANN pretrained as the encoder and a stack of two transformer decoder layers as decoder. To compare against large models, we show our audio captioning performance on HTSAT-BART[3] which consists of HTSAT as audio encoder and BART[3] as decoder. For the shared embedding space to get projections, we use CLAP[16] and HTSAT-BERT[3]. We use the LAION checkpoint[16] for CLAP and WavCaps checkpoint for HTSAT-BERT[3]. We use 0.8 and 0.6 as α value for experiments in Table 4 and Table 5. CLAP Beam and HTSAT-BERT Beam refers to the proposed faithful decoding algorithm with CLAP and HTSAT-BERT as the shared embedding space to project audio and text respectively.

5. RESULTS

From Table 4, the improvements on CLAPScore$_{tt}$ by 0.04 and 0.06 for AudioCaps and Clotho datasets indicate reduced hallucinations. We also observe that the proposed faithful decoding improves the performance of baseline model across all metrics except ROUGE L. This demonstrates that the proposed faithful decoding not only reduces hallucination but improves overall caption quality for smaller models. As a sanity check and to compare the performance against larger models, we perform the same experiments on HTSAT-BART[3]. In Table 5, the proposed faithful decoding slightly outperforms on SPICE and CLAPScore$_{tt}$ for AudioCaps. This is expected since the HTSAT-BART, being a large model and trained on Wavcaps[3] a much larger dataset (630k samples) than Clotho and AudioCaps, does not get extra useful information from the shared embedding space. The drop in CIDER scores for the proposed faithful decoding is due to CIDER not taking into account audio or text semantic similarities and instead relying only on word-frequency (i.e., TF-IDF).

6. CONCLUSION

We investigated the hallucination problem in audio captioning and proposed a new hallucination augmentation technique which will aid in future research of hallucination mitigation algorithms and metrics. Then, we showed that cosine similarity on audio-text shared embedding is a good hallucination metric. With no further finetuning, we proposed an inference time faithful decoding algorithm that utilizes shared embedding space to guide the beams during decoding time. In the future, we plan to develop a hallucination loss for finetuning stage.

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